

Student 2 - Java Concepts & Class Design

Time: 2 minutes 30 seconds

Slides: Slide 03 (Java Concepts Used) + Slide 04 (Class Design)

"Now I will explain the Java concepts used in our project."

Click to slide 03 - Java Concepts Used.

"Our project uses nine Java concepts. Each card shows the syntax used."

Point at the grid of 9 concept cards.

"First - Variables and data types. We use int id, String name, boolean isOpen."

Point at card 1 (STRING badge).

"Second - ArrayList. We use List of Item equals new ArrayList to store items."

Point at card 2 (LIST badge).

"Third - Static methods. We use static for isValidDate and searchBy methods."

Point at card 3 (STATIC badge).

"Fourth - Conditional statements. We use if else to check user input."

Point at card 4 (IF/ELSE badge).

"Fifth - Classes and objects. We have 6 classes. We create objects like new Item."

Point at card 5 (6 CLASSES badge). Emphasize 6 classes.

"Sixth - Encapsulation. We use private fields and public getters."

Point at card 6 (PRIVATE badge).

"Seventh - Loops. We use while true for the menu loop and for each for items."

Point at card 7 (WHILE/FOR badge).

"Eighth - Constructors. We use Item constructor with this dot id equals id."

| Point at card 8 (ITEM badge).

"Ninth - Streams and lambdas. We use items dot stream dot filter dot collect."

| Point at card 9 (STREAM badge).

"Now let me show you the class design."

| Click to slide 04 - Class Design.

"We have 6 classes in the UML diagram. Main, ItemService, StorageManager, Item, Validator, and Tests."

| Point at each class box in the UML diagram.

"Main handles the menu and user input. It has Scanner and ItemService."

| Point at Main class box.

"ItemService is the business logic. It has postItem, getLostItems, getFoundItems."

| Point at ItemService class box.

"StorageManager handles file operations. It has load, append, and rewrite."

| Point at StorageManager class box.

"Item is the data model with 8 fields - id, type, name, category, date, location, contact, status."

| Point at Item class box showing the fields.

"Validator is a utility class with isEmpty and isValidDate."

| Point at Validator class box.

"Tests has passed and failed counters and a check method."

| Point at Tests class box.

"Main never touches the file. ItemService never reads user input. Each class has one role."

| Trace the dependency arrows: Main uses ItemService, ItemService uses StorageManager, StorageManager reads writes data/items.txt.

"I will now hand over to explain the implementation."

| Gesture towards next presenter.